

Solvency Field Theory V12.6

Retarded Source, Cadence Holonomy,
and Closure Spectrum Scaffold

A checkpoint from toy Dent pushforward
through discrete spectral closure

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Interior cadence modes close by reflection; the high endpoint closes against the zero boundary.

Publication boundary. This checkpoint extends V12.5 by preserving the R61–R82 post-V12.5 gate sequence. It does not claim a derivation of the real particle spectrum. It shows that a retarded overlap source scaffold can be linked to proper-time cadence, discrete normal-bundle holonomy labels, a no-target spectral roster, reflection closure, and a zero-boundary endpoint candidate.

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1 The physical picture

Why do only specific particles exist?

The Standard Model lists the particles. Six quarks. Six leptons. Four force carriers. One Higgs boson. Seventeen entries on a table, each with a measured mass, and no explanation of why these masses and not others. The mass spectrum is the most precisely measured and least understood feature of the Standard Model. It is an answer with no derivation.

This document asks a structural version of that question: if a particle is a real maintained structure — a worldline tube processing at c with a definite Compton cadence — then are all cadences allowed, or does closure force a discrete set?

The answer from R72–R82 is that closure forces a discrete set. Not the measured set. Not yet. But the mechanism that produces discreteness has been identified and tested.

1.1 Why discreteness might be forced

A particle maintains itself by cycling. Each tick writes one Planck area to the boundary. The cadence — the rate of cycling — is set by the mass: $\omega_C = mc^2/\hbar$. But not every cadence may be self-consistent. The worldline tube has a normal bundle — the transverse directions perpendicular to the worldline. That bundle carries holonomy: as the worldline advances, the transverse frame rotates, and the rotation must close after one complete circuit.

Closure means: the frame must return to a consistent orientation. It need not return to *exactly* its starting angle — that would allow only integer winding. But it must return to an orientation that is compatible with continued maintenance. The allowed orientations are the modes of the normal bundle. Different modes correspond to different cadence ratios. And only a discrete set of ratios satisfies the closure condition.

This is the same mathematics that produces the discrete energy levels of a quantum system, the allowed harmonics of a vibrating sphere, and the quantized angular momentum values in atomic physics. Discreteness from closure is the most universal mechanism in physics. V12.6 asks whether it applies to particle cadences.

1.2 From overlap to source law

V12.5 established the Dent/Born pushforward: external interactions place phase-dependent “dents” on a particle’s trajectory through retarded potentials, and the Born rule is the pushforward of the canonical completion measure through that bounded dent map.

R61–R71 road-tested this machinery. The toy pushforward works structurally. Slit and screen geometries produce recognizable signal. But the raw phase gradient is dangerous near nodes — it diverges where the amplitude passes through zero. The current density $\mathbf{j} = \text{Im}(\psi^* \nabla \psi)$ is safer, and the endpoint PDE formulation is cleaner than direct phase tracking. Hard cores do not overlap (no-overwrite), but edge, framing, and current envelopes can — interactions happen through the overlap of maintained influence envelopes, not through contact.

1.3 From source to cadence

R72–R75 made the critical upgrade: replacing the toy coupling parameters with proper-time cadence. The Dent/Born source is not a free knob. It is the retarded overlap between maintained worldline tubes, propagated causally from the past light cone. The coupling strength is not a parameter — it is a cadence-memory response: how strongly one worldline’s maintenance history imprints on another’s completion space.

The key move is replacing the artificial toy phase with the proper-time Compton winding $\phi = \omega_C \tau$. The source becomes physical: the retarded dent amplitude ϵ_{ret} and orientation χ_{ret} are determined by the past interaction history, not by a free parameter. The cadence $\omega_C = mc^2/\hbar$ sets the time scale of the source oscillation.

1.4 From cadence to holonomy

R76–R78 asked: what cadence ratios are allowed? If the worldline’s normal bundle is compact — if the transverse frame lives on a circle — then the allowed winding modes are discrete. For a normal bundle with periodicity 2π , the allowed modes are labeled by integers m and N , giving cadence ratios $r = m/N$.

Not all ratios are equally costly. The simplest modes (small m , small N) have the lowest structural action cost. Complex modes (large m , large N) cost more to maintain. The action-cost ranking produces a natural ordering: the lightest, simplest modes are the ones most easily maintained.

The result is a scaffold, not a unique spectrum. The action-cost comparison is robust across multiple candidate action functionals (R78), but the unique action that selects exactly the physical spectrum has not been derived.

1.5 From holonomy to closure

R79–R82 asked the sharpest question: given the holonomy modes, which ones *close*? Two closure rules were tested.

Reciprocal closure: modes r and $1/r$ pair. This produces too many modes because modes like $r = 1$ are self-paired.

Reflection closure: modes pair under $r \mapsto 2 - r$. This is stronger — it pairs interior modes symmetrically and produces a tighter roster. The reflection rule has a natural interpretation: the total cadence range $[0, 2]$ is bounded, and modes are reflected against the midpoint $r = 1$.

But the reflection rule drops the endpoint: the mode $r = 2$ maps to $r = 0$, which is not an ordinary particle. R82 resolved this: $r = 0$ exists as a boundary anchor — a legitimate zero-mode object ($N = 1$, $m = 0$, $q = 0$) that terminates the spectrum without being a particle species. The mode $r = 2$ is admitted because it pairs with the boundary anchor, not with an interior mode.

The resulting roster of twelve modes:

$$\boxed{\frac{1}{2}, \frac{2}{3}, \frac{1}{3}, \frac{3}{4}, 1, \frac{1}{4}, \frac{3}{2}, 2, \frac{4}{3}, \frac{5}{3}, \frac{5}{4}, \frac{7}{4}} \quad (1)$$

Twelve discrete cadence ratios, from one closure rule, with no fitting.

1.6 What this is and what it is not

The twelve-mode roster is a *scaffold*. It demonstrates that closure conditions on the holonomy ladder produce a finite discrete spectrum rather than a continuum. It does *not* claim to be the real particle mass spectrum. The Standard Model masses are measured to many digits and the mapping from toy cadence ratios to physical masses requires the absolute cadence scale Ω_0 , envelope normalization, and pressure calibration — none of which are derived.

The value of V12.6 is structural: discreteness is forced by closure, not assumed. The mechanism is identified. The specific spectrum it produces remains an open research target.

2 Source-law road testing (R61–R71)

2.1 Toy Dent pushforward

R61–R64 road-tested the V12.5 Dent/Born machinery on slit and screen geometries. The toy pushforward produces recognizable signal: phase-dependent deflection through a slit creates a spread on the detector that depends on the slit width and the particle’s cadence. The structural result is that the machinery *works* — it produces output rather than diverging or collapsing.

2.2 Node and current refinements

R65–R69 exposed a critical technical issue: raw phase gradients diverge near amplitude nodes ($R \rightarrow 0$). The current density $\mathbf{j} = \text{Im}(\psi^* \nabla \psi)$ provides a node-safe replacement. The proper-time normalization requires careful treatment to avoid artifacts. The mobility- τ normalization (R69) produces well-behaved transport coefficients.

2.3 Overlap taxonomy

R70–R71 defined what can overlap and what cannot. Hard cores — the Planck-area write patches — cannot overlap (no-overwrite). But the extended influence envelopes of maintained worldlines can overlap at their edges. The edge/framing/current overlap is the physical source of inter-worldline interaction. Interactions happen through the overlap of maintained envelopes, not through hard-core contact.

This taxonomy connects to V12.4’s force picture: gravity is the scalar monopole overlap, electromagnetism is the framing/holonomy overlap, and the strong force may involve the topological overlap of bundled connections. Each force sector corresponds to a different channel of envelope overlap.

3 Retarded source and proper-time cadence (R72–R75)

3.1 Overlap to retarded source

R72–R73B converted the allowed envelope overlap into a retarded source map. The dent that worldline A places on worldline B propagates causally from the past light cone. The retarded source amplitude and orientation are determined by the history of A ’s maintenance activity as seen from B ’s current position:

$$U_{\text{dent}} = \epsilon_{\text{ret}}(x, t) J F(\phi - \chi_{\text{ret}}), \quad (2)$$

where ϵ_{ret} and χ_{ret} are evaluated on the past light cone. The causal-domain repair (R73B) ensures that only past-light-cone source data enters the interaction.

3.2 Proper-time cadence

R74 replaced the toy phase parameter with the proper-time Compton winding:

$$\phi(\tau) = \omega_C \tau = \frac{mc^2}{\hbar} \tau. \quad (3)$$

This is the particle’s own rest-frame clock, not an external parameter. The source oscillation is now physically grounded: it oscillates at the Compton frequency because the particle’s internal maintenance cycles at that rate.

3.3 Cadence-memory coupling

R75 partially replaced the free coupling strength with a cadence-memory response: the strength of the Dent interaction between two worldlines depends on the ratio of their cadences and the overlap geometry of their maintenance envelopes. This is not a derivation of the coupling — the full normalization remains open — but it constrains the coupling to depend on cadence rather than being a free parameter.

4 Discrete holonomy labels and action cost (R76–R78)

4.1 Normal-bundle holonomy

The worldline tube has $D_{\perp} = 4 - 1 = 3$ transverse directions in 3+1D spacetime. The transverse frame can carry winding modes on the normal bundle’s compact angular directions. For a circle of periodicity 2π , the modes are labeled by coprime integers:

$$r = \frac{m}{N}, \quad \gcd(m, N) = 1, \quad N \geq 1, \quad m \geq 0. \quad (4)$$

The cadence ratio r is the winding number of the transverse frame per unit of the base cadence Ω_0 . Different values of r correspond to different species — different internal structures maintaining themselves at different rates.

4.2 Action-cost ranking

R77 ranked the holonomy modes by structural action cost. Several candidate action functionals were tested (curvature-squared, winding length, entropy, complexity). All produce consistent low-cost rankings for the simplest modes. But R78 showed that the specific ranking is underdetermined: multiple action functionals agree on which modes are cheapest but disagree on the detailed ordering of intermediate modes.

The robust result: simple modes (small m , small N) are cheap. Complex modes (large m , large N) are expensive. The cheapest modes are the ones most likely to be maintained. The detailed spectrum requires a unique action functional, which remains open.

5 No-target spectral roster (R79–R80)

R79 built a spectral roster *without targeting* any known spectrum. The construction used only the holonomy labels, the action-cost ranking, and a maximum-complexity cutoff $N_{\max}$. The roster was frozen before any comparison to toy target values.

The roster showed predictive structure: it clusters modes at specific ratios that are not uniformly distributed, and the gaps between modes have a pattern related to the Farey sequence of rational numbers with bounded denominators.

R80 tested sharpening constraints — additional structural requirements (e.g., reflection symmetry, charge-conjugation compatibility) that reduce the roster. The sharpened roster retained the held-out toy modes and rejected several spurious candidates, showing that the structural constraints have real discriminating power.

6 Reflection closure and zero-boundary endpoint (R81–R82)

6.1 Reflection closure

R81 tested two closure rules on the spectral roster.

Reciprocal closure ($r \leftrightarrow 1/r$) pairs modes whose cadence ratios are multiplicative inverses. This is natural for oscillators: frequency and period are reciprocal. But it produces a loose roster with self-paired modes ($r = 1$) and no upper bound.

Reflection closure ($r \mapsto 2 - r$) pairs modes by reflection around $r = 1$ within the bounded range $[0, 2]$. This is stronger: every interior mode has a unique partner, and the spectrum is bounded. Reflection closure produced a tighter, more structured roster than reciprocal closure.

The structural interpretation: the total cadence range is $[0, 2]$ (twice the base cadence). Modes are symmetric under reflection about the midpoint. A mode at ratio r has a partner at $2 - r$. The spectrum is bounded and discrete.

6.2 The zero boundary anchor

The reflection rule sends $r = 2$ to $r = 0$. But $r = 0$ means zero cadence — no internal cycling, no mass, no maintenance. Is that a particle?

R82 showed that $r = 0$ is not a particle species. It is a boundary anchor: a legitimate zero-mode object with $N = 1$, $m = 0$, $q = 0$. It terminates the spectrum without participating as an ordinary species. The mode $r = 2$ is admitted because its reflection partner is the boundary, not another species.

Interior cadence modes close by reflection. The high endpoint closes against the zero boundary.

The resulting roster of twelve allowed cadence ratios:

$$\boxed{\frac{1}{2}, \frac{2}{3}, \frac{1}{3}, \frac{3}{4}, 1, \frac{1}{4}, \frac{3}{2}, 2, \frac{4}{3}, \frac{5}{3}, \frac{5}{4}, \frac{7}{4}} \quad (5)$$

Twelve modes. One closure rule. No fitting. No target.

7 What is earned, conditional, and open

7.1 Earned

A continuous scaffold from retarded overlap source through proper-time cadence to discrete holonomy modes and reflection-closed spectral roster.

Hard-core no-overlap, edge/envelope overlap taxonomy.

Node-safe current replacement for raw phase gradients.

Endpoint PDE formulation is cleaner than direct phase tracking.

Action-cost ranking is robust across candidate functionals.

Reflection closure is stronger than reciprocal closure.

Zero mode is a boundary anchor, not a species.

7.2 Conditional

Discrete holonomy spectrum if the normal-bundle compact periodicity and action functional are earned.

Twelve-mode roster if reflection closure is the native rule.

Zero-boundary endpoint if the native endpoint theorem is proven.

7.3 Open

Native retarded worldline source law.

Unique normal-bundle action functional.

Native endpoint theorem ($r = 0$ as boundary anchor from first principles).

Absolute cadence scale Ω_0 .

Envelope width and coupling normalization.

Mapping from toy cadence ratios to real particle masses.

Detector settlement and Born readout connection.

Real mass hierarchy.

7.4 Rejected routes

Raw phase gradient as source: rejected. Diverges at nodes.

Hard-core overlap as interaction mechanism: rejected. No-overwrite forbids hard-core overlap. Interactions happen through edge/envelope overlap.

Reciprocal closure as the native rule: rejected in favor of reflection closure, which is structurally stronger.

Endpoint $r = 2$ as an ordinary species: rejected. It is admitted only through the zero-boundary anchor mechanism.

8 Conclusion

V12.6 preserves a discrete closure-spectral scaffold. The scaffold demonstrates that normal-bundle holonomy, combined with reflection closure and a zero-boundary endpoint anchor, produces a finite discrete set of allowed cadence modes — twelve in the simplest implementation. This is a mechanism for discreteness, not a derivation of the physical spectrum.

The value of the scaffold is structural: it shows that particle species, in a maintained-worldline picture, cannot have arbitrary masses. Closure forces a discrete set. The specific set depends on the action functional, the bundle periodicity, and the absolute cadence scale — all of which remain open. But the discreteness itself is not optional. It is forced by closure.

This checkpoint derives no real particle spectrum, but it preserves a discrete closure-spectral scaffold.

The mechanism for discreteness is identified. The specific spectrum it produces remains the next frontier.

Acknowledgments

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A Gate rollup

Gate	Title	Verdict
R61–R64	Toy Dent pushforward	Pass: slit/screen pushforward produces recognizable signal. Structural machinery works.
R65–R69	Node, current, τ refinements	Pass: current is safer than phase gradient; proper-time normalization avoids artifacts.
R70	Endpoint PDE formulation	Pass: PDE formulation is cleaner than direct phase tracking.
R71	Core-edge-field overlap taxonomy	Pass: hard cores cannot overlap; edge/framing/current envelopes can.
R72	Overlap gradient to retarded Dent source	Toy pass: allowed envelope overlap becomes retarded source candidate.
R73–R73B	Retarded two-worldline source and causal repair	Pass: retarded source with lightcone/root validity repaired.
R74	Native helix phase origin	Pass: proper-time winding replaces toy phase parameter.
R75	Cadence-memory coupling	Partial pass: cadence-memory response replaces free strength knob. Full normalization open.
R76	Normal-bundle holonomy species cadence	Scaffold pass: cadence ratios become discrete holonomy labels. Selection theorem open.
R77	Mode action-cost ranking	Scaffold pass: structural action cost ranks modes consistently.
R78	Unique action derivation	Robustness pass: multiple actions agree on cheap modes. Unique action not derived.
R79	No-target spectral roster	Partial pass: no-target roster shows predictive structure. Unique spectrum open.
R80	Roster sharpening	Partial pass: sharpening constraints retain held-out modes and reject spurious candidates.
R81	Native closure constraint	Pass: reflection closure stronger than reciprocal closure.
R82	Zero-mode boundary endpoint	Partial pass: $r=0$ as boundary anchor admits $r=2$ endpoint cleanly. Native theorem open.